MILITARY SPECIFICATION

BARRIER MATERIAL, FLEXIBLE, WATERVAPORPROOF, PUNCTURE-RESISTANT, HEAT SEALABLE FOR PACKAGING AMMUNITION

1. SCOPE

1.1 Scope.-This specification covers flexible, watervaporproof, puncture-resistant, heat sealable barrier material for packaging ammunition items (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CCC-T-191	-	Textile Test Methods
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed,and Lock-Corner
PPP-B-636	-	Boxes, Fiber
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-D-723	-	Drums, Fiber
PPP-F-320	-	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Snapes
PPP-T-451	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	-	Tape: Pressure-Sensitive Adhesive, Filament Reinforced
TT-W-572	-	Wood Preservative, Water Repellent

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-P-116	-	Preservation, Methods of
MIL-P-46124	-	Plastic Molding and Extrusion Material, Ionomer Resins
MIL-S-4461	-	Sealing Machine, Heat, Hot Jaw and Continuous

FSC: 8135

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STANDARDS

FEDERAL

Fed. Test Method Std. No. 101 - Preservation. Packaging and Packing Materials Test Procedures

FED-STD-595 - Colors

Fed. Test Method Std. No. 601 - Rubber; Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATIONS

F-119-70 - Grease Penetration of Flexible Barrier Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

CONSOLIDATED FREIGHT CLASSIFICATION COMMITTEE PUBLICATIONS

Consolidated Freight Classification Ratings, Rules and Regulations

(Applications for copies of the above publications should be addressed to the Consolidated Freight Classification Committee, 202 Chicago Union Station, Chicago, Illinois, 60606).

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AMERICAN TRUCKING ASSOCATION, INC.

National Motor Freight Classification Rules and Container Regulations

(Applications for copies of the above publication should be addressed to the National Classification Board, 1616 P. Street, N.W., Washington, D. C., 20036).

3. REQUIREMENTS

3.1 Material.-The barrier material shall be constructed from the components listed in 3.2. It shall be constructed by such processes so as to comply with the requirements of this specification, and so that it can be fabricated into sleeves, bags and pouches by heat sealing the material to itself (top ply) (face side). (See 6.1).

3.2 Construction.-The barrier material shall be constructed as follows:

Top or Face Ply -	65 pounds per ream, (3,000 square feet), ionomer, conforming to specification MIL-P-46124, Type V, Class 2.
Second ply -	0.0007 dead soft aluminum foil
Third ply -	30 pounds per ream, ionomer conforming to Specification MIL-P-46124, Type V, Class 2
Fourth ply (Backing) -	1.6 ounce per square yard spun bonded high density polyethylene

3.3 Form.-The barrier material shall be furnished in roll or cut sheets form, as specified in the contract or order. Unless otherwise specified, rolls shall be at least 100 yards in length and 36 inches in width. The dimensions of flat cut sheets shall be as specified in the contract or order. Tolerances for cut sheets shall be plus or minus (±) 1/4 inch, or as specified in the procurement document. Edges of all rolls and sheets shall be evenly cut without selvage. Each component shall cover the entire surface area uniformly and evenly.

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3.3.1 Splices.-No roll shall contain more than three splices, and no single piece within a roll shall be less than 25 yards in length. Splices shall be permitted only in the direction perpendicular to the machine direction. Splices shall be evenly made, the entire width of the roll, and shall not come apart during the normal unwinding of the roll. Rolls containing splices shall be flagged at both ends of each splice with colored markers to indicate the splice location within the roll. Barrier material furnished in flat cut sheet form shall contain no splices.

3.4 Material identification.-The "backing" side (see 3.2) of the barrier material shall be marked to show the following: manufacturer's designation, date of manufacture (month and year), specification number and the notation "seal other side". The above information shall appear in continuous rows of constantly recurring symbols in the roll machine direction, from one end to the other with two ± 1/16 inch between repeated symbols. A complete row of markings shall appear once in the center of the roll width on rolls up to 36 inches wide, and twice within the width of rolls over 36 inches wide, one row near each edge. Flat cut sheets will be cut from rolls so marked. The symbols shall be applied by marking ink or other suitable fluids not deleterious to the barrier material. They shall be lusterless red, approximating in color chip number 31136 of FED-STD-595, be clearly legible, not more than 3/16 inch, or less than 1/16 inch in height. Each roll or package of sheets of barrier material shall include a tag secured to the roll core, or a sheet inserted in the package of sheets, with the sealing conditions as furnished by the manufacturer to be used as a guide by users for satisfactory heat sealing on rotary, thermal impulse, band and jaw type (as applicable), heat sealing equipment.

3.5 Physical properties.-The physical properties of the barrier material shall comply with the requirements of Table III when tested as required in Section 4 of this specification.

3.6 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship.-The barrier material shall be manufactured in accordance with the best commercial practices, and shall be suitable for the intended uses. It shall be of uniform construction, free from holes, tears, cuts, sharp creases, wrinkles and other defects that might impair its usefulness.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First Article Inspection (See 6.2)

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample, consisting of one roll, (sufficient in size to perform the specified tests) shall be selected from consecutively produced rolls which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity, as determined by the Government.

4.2.2 Inspections to be performed.-The finished sample shall be subjected by the Government to any or all of the examinations or tests specified in this specification.

4.2.3 Rejection.-If the finished sample fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component to comply with any of the stated requirements

4.3 Inspection Provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to

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exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process or specification, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements.

4.3.2 Lot size.-A quality inspection lot shall consist of either 50,000 square yards of material or all the material manufactured by the same process, from the same components, during one production run, whichever is the lesser. For the purpose of maintaining continuous testing, test samples shall be selected immediately following manufacture, or from material packed for shipment. The lot shall be divided into five equal segments. Five sample rolls shall be drawn from the lot for testing (e.g., one roll from each lot segment) and numbered. From each roll selected, a sample of six yards will be drawn for test purposes.

4.3.3 Testing

4.3.3.1 Sampling.-For the examination specified, the sampling plan for each of the defects covered in Table I shall be in accordance with MIL-STD-105, Inspection Level S-3, AQL 4.0 percent.

4.3.4 Inspection.-The barrier material shall be weighed, visually and dimensionally inspected, as applicable, for compliance with the necessary requirements and for workmanship. Equipment necessary for the performance of the inspections listed shall be in accordance with paragraph 4.3.7.

TABLE I

Classification of Defects

Category	Defects	Method of Inspection	Code No. (See 6.5)
Width, Roll (See 4.3.4.2)	Minor	Scale	01001
Length, Roll (See 4.3.4.2)	Minor	Scale	01002

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Width, Cut Sheet (See 4.3.4.2)	Minor	Scale	01003
Length, Cut Sheet (See 4.3.4.2)	Minor	Scale	01004
Thickness (See 4.3.4.2)	Minor	Scale	01005
Weight (See 4.3.4.4)	Minor	Balance	01006
Form(See 4.3.4.1)	Minor	Visual	01007
Workmanship (See 4.3.4.1)	Minor	Visual	01008
Construction (See 4.3.4.1)	Minor	Visual	01009
Cleanliness (See 4.3.4.1)	Minor	Visual	01010
Identification (See 4.3.4.1)	Minor	Visual	01011
Unwinding of roll (See 4.3.4.2)	Minor	Visual	01012
Identification - Tag or sheet - as applicable (See 4.3.4.2)	Minor	Visual	01013
Packaging (See 4.3.4.3)	Minor	Visual	01014
Packing (See 4.3.4.3)	Minor	Visual	01015
Marking (See 4.3.4.3)	Minor	Visual	01016

4.3.4.1 Examination of the end item for defects in appearance and workmanship.-The sample unit for this examination shall be expressed in units of square yards of barrier material. Samples for examination shall be selected in accordance with 4.3.3.1. Sufficient rolls shall be selected at random so that by examining approximately 6 square yards per roll, the required sample yardage will be obtained. For examination of sheets, samples shall be scored only once for each occurence within a square yard.

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Examine	Defect
Form (Check both sides of barrier material)	Not roll or flat cut, as specified
Cleanliness(1)	Not clean
Workmanship	Delamination Embrittlement Cut Tear Chafed spot Other
Construction	Not in accordance with 3.2 Not uniform - pinholed, blistered, wrinkled, folds, fisheyes, gel, foreign matter.
Identification	Not in machine direction Legend not as specified (See 3.4) Parallel rows over 18 inches apart (in 36-inch wide rolls) Dimensions not as required in 3.4 Marking color not as specified in 3.4.

(1) Defect does not apply to outer convolution of roll.

4.3.4.2 Examination of the end item for defects related to the roll or package of cut sheets.-The sample unit for this examination shall be expressed in units of rolls or packages of flat cut sheets. Samples for examination shall be selected in accordance with paragraph 4.3.3.1.

Examine	Defect
Width, roll	More than 36-1/4 inches Less than 35-3/4 inches
Length, roll	Less than 100 yards
Unwinding of rolls	Material blocked on roll causing material damage. Roll unevenly wound Roll telescoped Core missing or damaged

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Length, flat cut sheet Width, flat cut sheet	Not as specified by more than tolerance allowed in 3.3 or as allowed in contract
Identification tag or sheet (as applicable)	Missing, incorrect, illegible

4.3.4.3 Examination of preparation for delivery.-The lot size shall be expressed in units of shipping containers, (one drum per roll; and bundle of flat cut sheets, as applicable), and the sample unit shall be one shipping container, fully packed selected prior to the sealing or tying operation.

Examination	Defect
Packaging	Not level specified, not in accordance with contract or order. Flat cut sheets not unit package and wrapped in bundles, as specified; Fiberboard sheets omitted from top or bottom of stack, or not of sufficient size to protect cut sheets in package; Ties (tape, twine, rope) not applied as specified; Packaging material not as specified; Closures not in accordance with specified method, or material.
Packing	Not level specified; Not in accordance with contract or order; Rolls not in fiber drums as required; Arrangement or number of rolls or flat sheet bundles not in accordance with requirements as per container. Container material not as specified. Closure not as specified.

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Marking	Interior or exterior marking omitted, incorrect, illegible, or not a required. Exceeds requirements.
Weight	Exceeds requirements.

4.3.4.4 Source of material used for examination. The same rolls or packages of cut sheets shall be used for the examinations of 4.3.4.1 and 4.3.4.2.

4.3.5 Quality conformance tests.-The barrier material shall be tested for compliance with the following requirements: Testing shall be in accordance with MIL-STD-105, Inspection Level S-1, AQL 1.5 percent (See 4.3.4.1).

TABLE II

Classification of Defects

Category	Defect	Method of Inspection	Code No. (See 6.5)
Seam strength	Major	Test	02001
Watervapor transmission rate	Major	Test	02002
Temperature flex	Major	Test	02003
Resistance to puncture	Major	Test	02004
Breaking strength	Major	Test	02005
Resistance to Preservative	Major	Test	02006
Resistance to penetration	Major	Test	02007
Resistance to Blocking	Major	Test	02008

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Contact corrosivity	Major	Test	02009
Accelerated Aging	Major	Test	02010
Flexibility	Major	Test	02011
Water Resistance of Marking	Major	Test	02012

4.3.5.1 Sample selection for quality conformance testing.- Quality conformance test samples shall be selected using the following procedure:

TEST	TEST SAMPLE ROLL NUMBER (See 4.3.2)
Seam strength after accelerated aging	1
Watervapor transmission rate after low temperature flex	2 and 5
Resistance to puncture after accelerated aging	1 and 4
Breaking strength after accelerated aging	2 and 3
Construction	4
Resistance to preservatives	3
Resistance to penetration	5

TABLE III

Physical Properties

Property	Test Paragraph	Requirement
Accelerated aging	4.4.1	Material shall show no visual evidence of cracking, coating loss or delamination

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FLEXIBILITY		
At room temperature (as received) (after aging) (at low temperature) 0°F	4.4.2	Flexed specimens (3 conditions) shall show no visual evidence of delamination, cracking or ply separation.
WATER VAPOR TRANSMISSION RATE		
At room temperature (as received)	4.4.3	0.01 grams/100 sq. in./24 hrs (max.)
At room temperature (after aging)		0.01 grams/100 sq. in./24 hrs. (max.)
At low temperature (-0°F)		0.02 grams/100 sq. in./24 hrs. (max.)
RESISTANCE TO PUNCTURE		
(as received)	4.4.4	35 lbs. ave. (min.)
(after aging)		30 lbs. ave. (min.)
THICKNESS (CALIPER)	4.4.5	0.013 $\pm$ 0.005 inch
RESISTANCE TO BLOCKING	4.4.6	No blocking, delamination, rupture
CONTACT CORROSIVITY	4.4.7	No corrosion or pitting on plate in contact area in visual examination
RESISTANCE TO PRESERVATIVES	4.4.8	No evidence of leakage, delamination, swelling or embrittlement visible
SEAM STRENGTH		
(as received) at room temperature	4.4.9 and	Shall support a 10-pound weight without separation or delamination for a period of 1 hour
at 160°F	Table IV	Shall support a 10-ounce weight without separation or delamination for a period of 1 hour.

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Sealed and aged		
at room temperature		Shall support a 10-pound weight without separation or delamination for a period of 1 hour.
at 160°F		Shall support a 10-ounce weight without separation or delamination for a period of 1 hour.
Aged & then sealed		
at room temperature		Shall support a 10-pound weight without separation or delamination for a period of 1 hour.
at 160°F		Shall support a 10-ounce weight without separation or delamination for a period of 1 hour.
WATER RESISTANCE OF MARKING	4.4.10	Shall be legible after 2 hours of immersion
BREAKING STRENGTH		
as received	4.4.11	115 pounds, average (min.)
after aging		110 pounds, average (min.)
RESISTANCE TO GREASE (OIL) PENETRATION	4.4.12	No visual evidence of penetration, delamination or swelling.

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TABLE IV

SEAM STRENGTH

SEAM MATERIAL CONDITIONED AT	SEAM SPECIMENS TEST AT CONDITION	NUMBER TEST SEAMS	TEST WT PER/SEAM	TEST TIME INTERVAL	TESTED SEAMS TO BE EXAMINED FOR
Room Temperature 1 hour at 73 ± 2°F and 50 ± 5° R.H.	(A) Room Temp (73 ± 2°F)	2	10 lbs.	1 hour	any separation of ply
	(B) + 158° ± 2°F	2	10 ozs.	1 hour	delamination
Seams fabricated from material as received and then subjected to paragraph 4.4.1.(Accelerated Aging Procedure)	DITTO	2 2	10 lbs. 10 ozs.	1 hour 1 hour	DITTO
Seams fabricated from material previously subjected to paragraph 4.4.1. (Accelerated Aging Procedure)	DITTO	2 2	10 lbs. 10 ozs.	1 hour 1 hour	DITTO

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4.3.6 Sealing instructions for first article inspection and quality conformance testing.-Heat seals for test purposes shall be a minimum of one (1) inch in width and shall be fabricated using a jaw type heat sealing machine conforming to Military Specification MIL-S-4461, with one heated jaw and one resilient unheated jaw. The sealing conditions shall be those recommended by the material supplier. The upper limits for the heat sealing conditions for this type of sealer shall not exceed 525°F, a 4-second dwell time interval, and an effective jaw pressure of 65 pounds per square inch (psi).

4.3.6.1 Test conditions.-Unless otherwise specified, the physical testing under this specification shall be made under atmospheric conditions of 73 ± 2°F, and 50 ± 2 percent relative humidity. The material shall be considered in equilibrium after 24 hours exposure to the specified atmosphere.

4.3.7 Inspection equipment.-For the performance of all tests and examinations specified in 4.3 and 4.4, commercial inspection equipment should be employed (see 6.4). The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of insuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes.

4.4 Test methods and procedures

4.4.1 Accelerated aging procedure.-Barrier material comprising approximately five square yards shall be subjected to conditioning at 158 ± 2°F within a circulating air oven or chamber for a period of two weeks (336 hours). The material shall be folded loosely, hung, rolled loosely, or laid flat in the oven. Following the above time interval, the material shall be returned to room temperature, and examined visually for any and all of the applicable defects of Table III. If the material is satisfactory, it shall be retained for use as "aged" material for the testing required elsewhere in this specification.

4.4.2 Flexibility procedure.-The flexing apparatus, test specimen preparation and flexing procedure shall be in accordance with Federal Test Method Standard No. 101, Method #2017. The full stroke operation (6 inch) shall be used. Four specimens, two in each major direction (machine and cross), shall be flexed in the three conditions as given in Table III. If, upon visual examination, flexed specimens are satisfactory, they shall be retained for use as test specimens, for 4.4.3.

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4.4.3 Water vapor transmission rate (WVTR) Method A.-The flexed sleeves retained from 4.4.2 shall be fabricated into pouches by heat sealing one open end, as specified in 4.3.6. Approximately 50 grams of calcium chloride contained in a heat sealable non-woven porous bag (see 6.3), is placed inside each pouch, which is then closed on the remaining open end by heat-sealing as before. Each pouch shall be conditioned for 16 hours at 100 ± 2°F and 90 ± 5% relative humidity (RH), and with negligable air circulation. After conditioning, the pouch shall be opened by carefully cutting off one heat sealed end, the calcium chloride removed, and a freshly weighed calcium chloride bag inserted and the bag sealed, as before. The resealed pouch and contents shall be exposed for a period of 66 ± 2 hours in the heat-humidity cabinet, as above; removed, exposed to conditioning of 4.3.6.1 for 2 hours, repoened and the calcium chloride reweighed. All weighing shall be to the nearest milligram (mg).

Method B (Alternative Method).-Infrared diffusometer method.- Alternately, if the equipment is available, the infrared diffusometer may be used to determine the moisture vapor transmission rate (MVTR) of the barrier material. The IRD-2 model is acceptable (or approved equal), for these determinations. The test is made using suitable specimens cut from the flexed sleeves as given in Method A, above. The operating setting of 100°F is to be used. (See 6.4.3).

4.4.3.1 Water vapor transmission rate calculations (For Method A).-The water vapor transmission rate shall be reported in grams of water vapor transmitted per 100 square inches of material area per 24 hours, calculated as follows:

$$WVTR = \frac{W2 - W1 \times 2400}{T \times A}$$

Where:

W1 = weight in grams of the calcium chloride at the start of the exposure period
W2 = weight in grams of the calcium chloride at the end of the exposure period
T = exposed period/hours (66 ± 2 hours)
A = Area exposed in square inches (inside bag surfaces).

Results for individual specimens shall be reported.

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4.4.3.1.1 Water Vapor Transmission Rate Calculations (For Method B).-Results obtained using the infrared diffusometer shall be reported in grams per 100 square inches per 24 hours. The values obtained using the above equipment are converted into the desired units (of grams, etc.) by conversion table supplied with the above equipment.

4.4.4 Resistance to puncture.-Puncture resistance shall be determined in accordance with Federal Test Method Standard No. 101, Method 2031, except that the "aged" specimens shall be cut from material from 4.4.1. Ten specimens each, as received, and aged (accelerated), shall be tested. The average obtained for each condition shall meet the applicable requirements of Table III.

4.4.5 Thickness.-Thickness (caliper) of the barrier material shall be determined in accordance with Fed. Test Method Std. No. 101b, Method 1003. Ten (10) random determinations shall be taken from as received material. Values shall comply with the applicable requirements of Table III.

4.4.6 Resistance to blocking.-Testing shall be in accordance with Procedure D of Method No. 3003, Federal Test Method Standard No. 101. The three sides of the test pouch shall be heat sealed under the conditions of 4.3.6. The resilient pads shall be composed of rubber having a type "A" Durometer hardness of 60. (See Federal Test Method Standard 601.)

4.4.7 Contact corrosivity.-Testing shall be in accordance with Method No. 3005, Federal Test Method Standard No. 101. Three specimens shall be tested. Following the test, if corrosion is visually evident in the blank area, the test shall be repeated using a new test panel. Results shall be reported on each material in triplicate. The test shall be repeated if one of the three specimens shows light or incipient corrosion. In the repeated test, all three specimens shall be satisfactory.

4.4.8 Resistance to preservatives.-Two 2 x 3 inch heat sealed pouches shall be formed of the material using the procedure of 4.3.6. Each pouch shall be partially filled (approximately 1/3) with clean dry shredded paper prior to sealing the fluid side of pouch. The sealed pouches shall be inserted into a covered desiccator having the bottom portion filled with preservative meeting Specification TT-W-572 for a period of 72 hours min. Reverse and examine for delamination, oozing, tacky surface, or blistering (See 6.6).

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4.4.9 Seam Strength

4.4.9.1 Seam specimen preparation.-For each condition listed in Tables III and IV and in 4.4.9.2, cut two 5 x 6 inch specimens of barrier material. Fold the specimens, heat seal face to heat seal face, to obtain specimens each 2 1/2 x 6 inches. Heat seal the opposite edge in accordance with 4.3.6 and cut open the fold. From the 2 1/2 x 6 inch specimen, cut 6-inch wide seam test specimens perpendicular to the seam, and discard the outer two. The remaining four 1 x 2 1/2 inch fin type seam specimens are for the specific test phase being conducted.

4.4.9.2 Test procedure.-After required conditioning, (see Tables III and IV; also paragraphs 4.3.6.1 and 4.4.1) each specimen to be tested for seam strength shall have one "leg" clamped so that the other leg hangs freely. Gently, so as not to impact, load each specimen by attaching applicable weights (the total weight for each condition (either 10 lbs or 10 ozs.) shall include the weight of the clamp. Following the one-hour test period at the required condition and load, the clamp and weight shall be removed and the seam specimens examined visually for evidence of separation or delamination. Any separation or delamination over 1/32 inch shall be cause for rejection (see Table IV).

4.4.10 Water resistance of marking.-Three specimens of appropriate size, each containing a set of markings (see 3.4) on the backing, shall be completely immersed in water at room temperature for two hours. The specimens shall be then removed, and the excess water removed from them by "blotting" with absorbent towels. The marking on the specimen backings shall be clear and legible.

4.4.11 Breaking strength.-Breaking strength shall be determined in accordance with Method No. 2037, Federal Test Method Standard No. 101 (also see CCC-T-191, Method 5100), Grab Method. Tests shall be conducted and the average breaking strength reported for 5 specimens in each principal direction (M and C), on both as received specimens and specimens cut from material "aged" in accordance with 4.4.1. The values obtained shall comply with applicable requirements of Table III.

4.4.12 Resistance to grease (oil) penetration.-Resistance to grease (oil) penetration shall be determined in accordance with ASTM-F-119 except that (a) hydraulic fluid meeting the requirements of Military Specification MIL-M-5606 shall be used as reagent, (b) an oven temperature of 60 degrees (°) + 1° Centigrade (C) shall be used, and (c) the reagent shall be applied to the backing side (tyvek or outer side) of specimen.

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5. PREPARATION FOR DELIVERY

5.1 Application.-The packaging, packing and marking requirements specified herein apply only to direct purchases by, or direct shipment to, the Government.

5.2 Packaging.-Packaging shall be Level A or C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Rolls and flat cut sheets.-Both rolls and flat cut sheets shall be packaged in accordance with MIL-P-116, Method 111, and as detailed below.

5.2.1.2 Rolls.-Each roll shall be wound on a substantial no-return fiber core with a minimum inside diameter of 3 inches + 1/8 inch, and shall be suitably restrained from unwinding during handling.

5.2.1.3 Flat cut sheets.-Cut sheets shall be sandwiched between two fiberboard sheets conforming to PPP-F-320, with a minimum bursting strength (dry) of 275 pounds. The flat cut sheets shall then be packaged in bundles having a maximum weight of fifty pounds. Bundles shall be secured with tape conforming to PPP-T-97, fiber twine or rope, two ties in each direction, of such strength as to secure safe arrival of the bundles. The fiberboard sheets shall be of a size commensurate with the dimensions of the flat cut sheets, so as to prevent damage to the barrier material during the bundling operations.

5.2.2 Level C.-Rolls and flat cut sheets shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his own standard practice when it meets the above requirements.

5.3 Packing.-Barrier material shall be packed Level A, B or C, as specified (see 6.2).

5.3.1 Level A

5.3.1.1 Rolls.-Each roll of material packaged as specified in 5.2 shall be packed in fiber drums conforming to Type III, Grade A, PPP-D-723. Drum closure shall be reinforced by means of three-inch wide tape conforming to PPP-T-451.

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5.3.1.2 Flat cut sheets.-Unless otherwise specified, one bundle of flat cut sheets, packaged as specified in 5.2.1.3, shall be packed in weather-resistant boxes conforming to PPP-B-636. The grade of box shall be governed by the size and weight limitation of the container specification. Four boxes shall then be overpacked in boxes conforming to PPP-B-601 (overseas type), PPP-B-621, or PPP-B-640, Class 2. End sheets of PPP-B-621 boxes shall be a minimum of 1 1/16 inches thick.

5.3.2 Level B

5.3.2.1 Rolls.-Each roll of material, packaged as specified in 5.2, shall be packed in fiber drums conforming to Type 1, Grade A of PPP-D-723. The covers shall be taped to the side walls with 3-inch usual reinforced tape or equivalent.

5.3.2.2 Flat cut sheets.-Each bundle of flat cut sheets, packaged as specified in 5.2 shall be packed in fiberboard boxes conforming to Type SF, class domestic of PPP-B-636. When specified (see 6.2) V3c or V3s boxes of PPP-B-636 shall be used.

5.3.3 Level C.-Rolls and flat cut sheets shall be packed in exterior shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall conform to Uniform Freight Classification Rules or to rules of other common carriers applicable to the mode of transportation.

5.4 Marking.-All individual packages and shipping containers shall be marked in accordance with MIL-STD-129, and as follows:

Size - Nominal net lineal yardage of roll or dimensions of flat cut sheets (net lineal yardage is the number of yards of usable material in the rolls), date of manufacture (month & year) " KEEP COOL AND DRY".

6. NOTES

6.1 Intended use.-It is intended that this barrier material be used primarily for the fabrication of bags, pouches and sleeves for packaging army ammunition items; where its properties, particularly its low water vapor transmission rate, high puncture resistance and breaking strength are important. Material having the constructions given in 3.3 has been found to be satisfactory.

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6.2 Ordering data.-Requests, requisitions, schedules and contracts or orders should contain the following:

a. Title, number and date of this specification
b. Quantity (Specify width and length for flat cut sheets (see 3.4).
c. Form
d. Provisions for submission of first article samples.
e. Weather-resistant class boxes if required for Level B (See 5.3.2.2).
f. Level of packaging and packing (See 5.2 and 5.3).

6.3 Test procedure material sources.-Sources of materials vital in test procedures are as follows:

a. Material for fabrication of desiccant bags for use in 4.4.3 can be obtained from:

(1) Webril Cloth - The Wendall Company, 309 West Jackson Boulevard, Chicago, Illinois.

(2) Lantuck, Wellington Sears Company, New York, New York, and Chicopee Mills; Massachusetts.

b. The petroleum base oil of 4.4.8 may be obtained from: Sun Oil Company, Special Products Division, 1616 Walnut Street, Philadelphia, Pennsylvania.

6.4 Inspection Equipment

6.4.1 Commercial inspection equipment is defined in AMC Regulation 702-2.

6.4.2 Contractor-designed equipment.-In the event that a contractor elects to design his own inspection equipment, details of the designs (drawings, description, materials, etc.) shall be submitted to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-P, Dover, New Jersey, 07801, for approval prior to fabrication and use. Approval of such designs may be delegated to the contract administration office for minor defects only.

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6.4.3 Equipment sources.-Sources of equipment cited in this specification are:

a. The Naval Air Engineering Center, Aeronautical Materials Laboratory Flex-tester (Gelbo-Tester) cited in 4.4.3 may be obtained from: The Atmosphere Control Company, Inc., 5315 Chester Avenue, Philadelphia, Pennsylvania.

b. Infra-red Diffusometer - may be obtained from: Modern Controls, Inc., Minneapolis, Minnesota.

6.5 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.6 Resistance to preservatives.-For the performance of the test specified in 4.4.8, an explosive proof or steam heated oven should be used. In addition, extreme caution should be exercised in handling Pentachlorophenol solution subjected to temperatures in the vicinity of 100°F (flashpoint of solution).

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Preparing Activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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